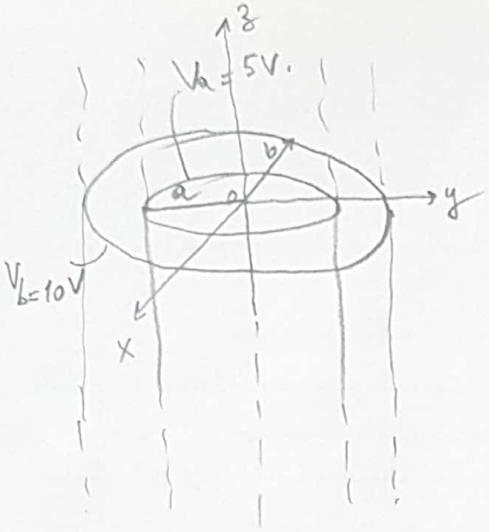
1.

a)



i) $0 \le \rho = \sqrt{x^2+y^2} \le a \Rightarrow V = V_a = 5\,V$

ii) $a \le \rho = \sqrt{x^2+y^2} \le b \Rightarrow \Delta V = \Delta V(\rho) = \frac{d}{d\rho}\left(\rho \frac{dV}{d\rho}\right) = 0 \Rightarrow \rho \frac{dV}{d\rho} = A = sbt$

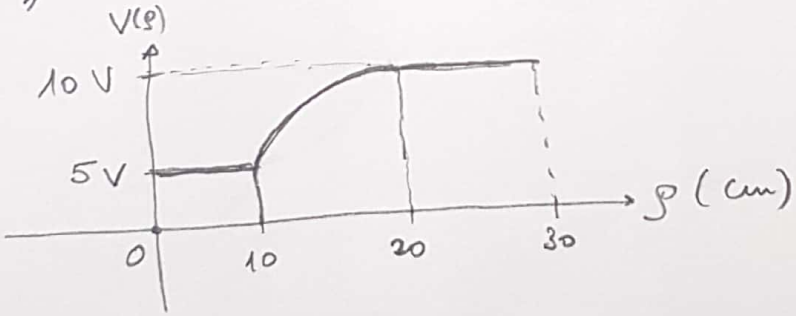
$$\Rightarrow \frac{dV}{d\rho} = \frac{A}{\rho} \Rightarrow V = \int \frac{A}{\rho} d\rho + B = A \ln \rho + B$$

$V_a = V|_{\rho=a} \Rightarrow A\ln a + B = V_a = 5\,V.$ $\quad V_b = V|_{\rho=b} \Rightarrow A\ln b + B = V_b = 10\,V$

$$\Rightarrow A[\ln b - \ln a] = 10 - 5 = 5V \Rightarrow A = \frac{5}{\ln(b/a)} = \frac{5}{\ln(20/10)} = \frac{5}{\ln 2}$$

$$B = V_a - A\ln a = V_b - A\ln b = 5 - \frac{5}{\ln 2}\cdot \ln(0{,}2) = 10 - \frac{5}{\ln 2}\cdot \ln(0{,}2)$$

iii) $\rho = \sqrt{x^2+y^2} \ge b \Rightarrow V = V_b = 10\,V$

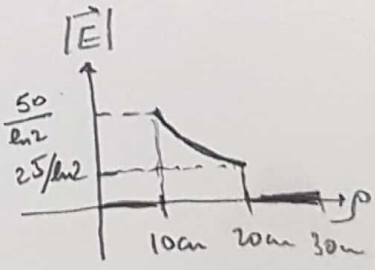


b) $$\vec{E} = -\text{grad}V = -\text{grad}\begin{cases} 5V & 0 \le \rho \le a \\ A\ln\rho + B & a \le \rho < b \\ 10\,V & \rho \ge b \end{cases} = -\begin{cases} \text{grad}\,5 & 0 \le \rho \le a \\ \text{grad}(A\ln\rho + B) & a \le \rho \le b \\ \text{grad}\,10 & \rho \ge b \end{cases}$$

$$-\vec{e}_\rho \cdot \underbrace{(A\ln a + B - 5)}_{0}\delta(\rho - a)$$

$$-\vec{e}_\rho \cdot \underbrace{(10 - A\ln b - B)}_{0}\delta(\rho - b)$$

$$\vec{E} = \begin{cases} 0\,; & 0 \le \rho \le a \\ -\frac{A}{\rho}\vec{e}_\rho & a \le \rho \le b \\ 0\,; & \rho \ge b \end{cases} \Rightarrow |\vec{E}| = \begin{cases} 0\,; & 0 \le \rho \le a \\ \frac{|A|}{\rho}\,; & a \le \rho \le b \\ 0\,; & \rho \ge b \end{cases}$$



CamScanner ile tarandı